

THE KERMOND CERTIFICATE OF EDUCATION

STUDENT NAME: _____

Teacher Code : HEWP XIAR **KERJ** SOTS FIRR **ALIY** WANK ~~**RIBS**~~
SHAP

Mathematical Methods 3/4 APPLICATION TASK 1A

Question and Answer Book

The Adventures of William Patel (SAC-MAN)



The Movie: *The SAC Robbery-Part I*

Reading time is **15 Minutes**

Writing time is **120 Minutes**

Approved Materials:

- Any aids for the sketching of graphs
- A physical **OR** digital Mathematica Bound Reference
- **OR** A CAS Calculator with **physical** bound reference
- Stationary including pencils and a scientific calculator (if desired)

Materials supplied:

- Question and Answer book of k amount of pages
-

“THE SAC HAS BEEN STOLEN!!!!!!”

Greetings students, teachers and fellow members. I urge you to heed this message with great haste, a matter of national security has occurred, VCE Mathematical Methods SAC I has been stolen at 12:00 AM GMT + 11.

The SAC was notoriously convoluted and many students had feared it's prevalence. So succulent yet were the boons of knowledge bestowed upon by this most ybeautiful SAC, yet so magnificent in it's tender paper layout, imprinted upon each page an index to show students where they were. Yet such auspicious formatting of the mathematical equations in the codified syntax of LaTeX. Nothing in the world, not even my orchard of blackberries were as beautiful as this SAC.

John Kermond was the head of the JKSS math's department (As indicated by the name of the school) and he had spent hours everyday with caffeine fuelled nights, in which he knew not it was his dopamine fuelling him or pure epinephrine. Each day in which he had to balance with his gym routine lifting weights that could induce seismic waves in the earth.

Yet, the SAC, on the day on which it were to be administrated to the gloomy polish-deprived tables of students, it, it WAS STOLEN!!!, right there out of plain sight. Kermond could not believe his eyes. It was if it disappeared with the gust of the world

It is a most indubitably repugnant matter and one of national safety in the country. JKSS has never seen such a breach of academic integrity in it's years of existence, where I first cut the ribbon at the automatic door.

In response to this urgent issue, head of maths Mr. Kermond has, with my permission formed a search party to investigate and hopefully find the assessment once again, the following individuals will be part of this search party

John L. Kermond - Victoria's most renowned mathematics educator

William (SAC-MAN) Patel- A Student selected by Mr. Kermond to be in the party

Agent Finchy- An JKSS software engineer from the UK

I regret to inform you of this tragedy, JKSS will be entering lockdown mode till the SAC is recovered, all students are dismissed at lunchtime

Principal

SECTION A:

- Answer **all** questions in the booklet
 - All written responses need to be in English
 - If not specified then provide **exact** values for each question
 - Mathematica /CAS Syntax is **not** suitable as working out
 - Show all mathematical reasoning if a question is of multiple marks
 - The sections in italics are story only and not required for the assessment
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Question 1 (25 Marks)

MISSION STATUS: ACTIVATED

TIME: 8:40 A.M. (GMT+11)

WILLIAM PATELS HOUSE

William Patel didnt hear the PA speech as he was sick on the day. He is now at home watching his favourite TV channel, Channel 67 The Mathselorette, his eyes are glued to the screen ,watching Jim Kermond be given the final rose by Elise breslie. William is intrigued.

Suddenly he receives a call from his Methods teacher , Mr. Kermond

“PATEL, didnt you hear the PA announcement, you are part of the team to to recover the Methods SAC I”

“Umm wha-” Will shaked with aguish fear

“No time to tell you this now, come to school now , we NEED TO TALK in person”

“OK Mr. Kermond!!!”

“Oh yeah and be careful , our enemies may be watching.....”

William Patel jumps and sprints to JKSS as fast as his legs can move , William is able to model his own path as travelling through an x-y co-ordinate plane, let william’s house be $(x-2)$ metres east and y^2+5y+9 metres north of the Origin.

a.) State the distance of William’s House to the point $(0,0)$ in terms of x and y (1 Mark)

It is known the variables x and y is part of the family of simultaneous equations

$$ky+4x=9$$

$$-3y+(k-8)x=10$$

b.) Show that using algebra , that the equations have no solutions for $k=2$ and $k=n$,
Where n is an integer (2 Marks)

William figures $k=5$

It is known that on this x-y plane, The JKSS building is located on the point (x_1, y_1) , $y_1 > 0$
 The x-coordinate of the midpoint of William's House and JKSS is 3 and the ordinate is such
 that the distance between Will's House and JKSS is $\sqrt{7569}$ metres

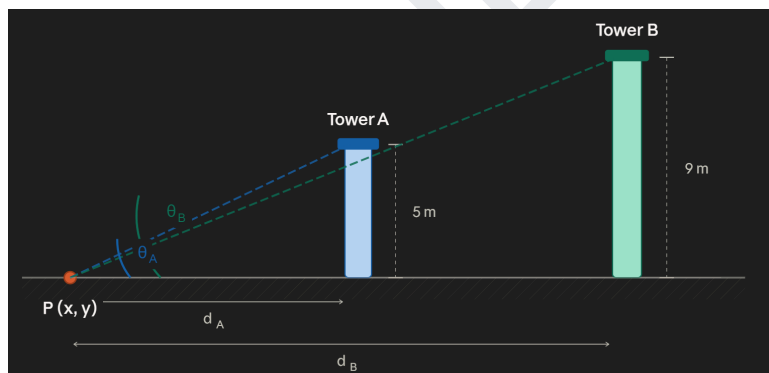
c.)

i.) State the co-ordinates of Will's House (1 Mark)

ii.) State the co-ordinates of JKSS correct to 2 decimal places (1 Mark)

iii.) Find the minimum distance between Will's House and the function $f(x)=x^2-2x+1$, correct to 3 decimal points (1 Mark)

William reaches the Midpoint $P(x, y)$ of his house and JKSS , he sees two towers based on the diagram shown . Tower A has a distance of 5 m and B with a distance of 9 m, The horizontal distance to the foot of Tower B is d metres and A and B have a horizontal distance of 5 metres from each other. let d be a distance from in between P and Tower A



d.)

i.) State a function in terms of the distance between Will and Tower A $\theta(d)$ that represents the difference in his viewing angles of the two towers (2 Marks)

ii.) For what co-ordinate of the base of Tower A is the viewing angle between the two lines of sight maximized

(2 Marks)

William runs at a velocity of 3ms^{-1} from his house at a bearing of 65 degrees True , when he observes a strange entity hovering in the air

Of a mechanical nature, it looked to be a drone with it's metal exoskeleton painted to be "ALBD-AI"

"ENTITY IDENTIFIED: WILLIAM PATEL

ACTION:**ATTACK**

ALBD-AI fires a laser beam that attempts to hit Will as he runs, it ends up bisecting the midpoint of the line connecting william and JKSS at $t=3$ seconds

e.) Find the equation of the laser beam in the form $y/m - x = c$ (2 Marks)

"NO , HOW DID HE ESCAPE" Uttered ALBD-AI with the might of it's training.

ALBD recalculates the gradient of the laser beam in order to hit william, it takes 2 more seconds for the beam to reach the line that which william is travelling on.

f.)

i.)

Determine the angle subtended by the line of the next laser beam, and the line that which william travels at that point , give your answer to three decimal places in **degrees**

(2 Marks)

ii.) What would be the gradient(s) of the laser beam if the angle subtended by it and Wil's path after 2 seconds be if the acute angle subtended by the lines was 50 degrees.

(2 Marks)

In order to avoid contact with the laser : William does an instinctive move, he observes a pool, William knows that due to training overfitting, ALBD's optical sensors are confused by the element of Chlorine, William can swim at a speed of $X1\text{ms}^{-1}$ and run at a speed of $X2\text{ms}^{-1}$ across the horizontal perimeter of the pool , a diagram of the pool is shown

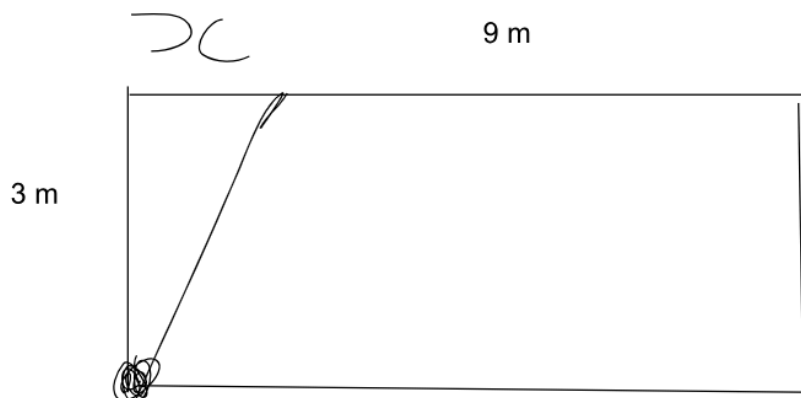


Figure 1. The pool William must cross to escape ALBD

William must swim through the pool till he reaches a point on the other side when he runs.

g.)

i.) William's run velocity is a metres, and his swim velocity is $X1/(10X2) \text{ ms}^{-1}$. $X1$ is the tens digit of swim velocity of the Channel Number of the Mathselorette and $X2$ is the ones digit, that he watched, state the family of quadratic expressions that has two solutions which are the two digits of the channel number (1 Mark)

ii.) State the **Distance** that William will travel across the pool in terms of x

(1 Mark)

iii.) State the **Time** that William takes to reach the end of the pool in terms of x (1 Mark)

If the minimum time taken to go through the pool is under 10 seconds William can escape ALBD-AI's laser, however the rate of change of time taken with respect to x must change within the motion

State the value(s) of a such that William can escape
(3 Marks)

ALBD-AI accidentally crashes into a tree before shooting the laser as it's optical sensors were confused by the presence of CHLORINE
"MAYDAY " Before a thunderous vociferation of sound waves went through the earth

"Phew" William Says as he makes it to the University into the Automatic Gates of JKSS
"WILL , ARE YOU OK " Asked Mr. Kermond

"Im fine sir, but something attacked me!" William told him breathless
"ALBD-AI, A Neural Network trained by our Nemesis , The Schmiddler"
"THE SCHMIDDLER!!!!!"? !? !? !? !? !? !"

"Hey Mr. Kermond, we gotta urgent report" - Shouted Agent Mann
"Speak "- Kermond responded hastily
"Open the news"
The news displays that an escape from Sharma Asylum resulted in approximately 10 casualties , a wanted prisoner is on the loose

"Me and Grant have analyzed the path of the train stations from that area I think we found the Schmiddler"
"What's Schmiddler's status"
"He's heading on the Pakenham line to Mt. Grainger"- Grant Chimed
"Mt. Grainger, why is he going there?"
"Wait Sir", said William "I think I know why, I remember the JKSS VCE Geography lesson plans involved analysis of the Mining activities in Mt. Grainger"
"Oh Mann...." Agent Gavrilward looked upon

Suddenly all TV Screens in the building turn to radio static

Kermond University's Electrical generator was disabled

ACTIVATING EMERGENCY SYSTEMS - AN AUTOMATIC VOICE HISSED**On the TV Monitor a set of co-ordinates briefly displayed for a few seconds****WE NEED TO GO NOW WILLIAM-MR. KERMOND SHOUTED****Question 2 (22 Marks) (T-Z)****MISSION STATUS: ONGOING****TIME: 12:00 PM (GMT+11)****The Lucas Leap highway (EN-ROUTE) TO MT. GRAINGER**

Mr. Kermond and Will get into Mr. Kermond's Self-driving Ferrari.

They meet Agent Finchy, a retired Software engineer who formerly worked for British Military Intelligence at the garage, getting out of his 1960s Chevy Impala.

They begin travelling across the Lucas Leap Highway **at a speed of 25 ms^{-1}**

" 12:00 AEST , TRAVELLING TO MT. GRAINGER VIA Lucas Leap , OVER AND OUT"-Mr. Kermond logged

The Lucas leap highway is situated just over an old closed down radioactive mine region, referred to as the Mt. Grainger radioactive zone

Mr. Kermond believes that the equation of Section A of the highway's perimeter can be modelled by the equation:

$$f: [0, 6] \rightarrow R, f(x) = ax^3 + bx^2 + cx + d$$

Where f is the height in hundreds of meters from the Radioactive zone and x is the horizontal distance in hundreds of meters from an origin point

It is known that at the beginning of the Lucas Leap highway, the car is 124200/50 metres above the radioactive zone , and touches it briefly after 300 metres and contacts the zone again just before transferring to section B,

a.) State the rule of $f(x)$,hence state the first time (XX:XX format) when Mr. Kermond's Ferrari has covered more than 10 metres of distance (2 Marks)

b.)

Mr. Kermond is not sure if his calculations are accurate, and analyzes the highway for more information

i.) **Find** the co-ordinates of the stationary points of $f(x)$ using the values obtained in part a. and state their nature (2 Marks)

William suggests translating $f(x)$ by a parameter k in the positive direction of the x axis,

ii.) **State** the value(s) of k for which $f(x)$ has one positive x intercept (1 Mark)

iii.) **For** what values of $f(x)-k$ are there two positive solutions (1 Mark)

Upon further inspection, it is revealed that Section A can be modelled by another function. Mr. Kermond suggests a model in the form

$$f(x) = Ax^3 + 2Ax^2 + 3x + 3$$

"Hmmm. how can we know this"

"Wait I know " Said William

When the **horizontal** distance of the stationary points of the highway, is less than 3 m ,
The Automated system gets confused and breaks

c.)

i.) **Find** the values of A for which Mr. Kermond would need to manually drive the ferrari (1 Mark)

,

He quickly realizes that the ferrari continues to be driven in the way it usually is (automatically)

“Ahh you bonehead!!!, Mr. Kermond screamed to himself”

He sees a hill yonder and believes the section B of the highway is modelled using:

$$f_2(x) = ax^5 - bx^2 + 1$$

ii.) Find the values of a and b such that the highway joins smoothly across the sections (1 Mark)

Mr. Kermond figures that $a = 1$ (it doesn't join smoothly apparently)

Section B of the Lucas Leap highway is modelled by

$$f_2(x) = \ln(f_2''(x))$$

Another road intersects Section B and is modelled by the function

$$g(x) = f_2^{-1}(3x)$$

ii.) For what value(s) of b to 2 decimal places do the roads intersect at least once (2 Marks)

Since the car does not need to operate manually, he deems the original function good enough

Some **goons** are tailing them on a truck behind following the path of $f(x)$, the truck has an automated lining system that creates a bridge from their position $(g, f(g))$, the bridge is tangent to the graph of $f(x)$ at g , $g < 4$ and makes an angle of 50 degrees with the x axis

ii.) Find the value of g (1 Mark)

iii.) If the graph of $f(x)$ is dilated by a factor of k from the x axis, $k > 0$, for what value(s) of k would the tangent line at $x = g$ intersect with f once, (2 decimal places) (1 Mark)

iv.) If it was translated $23/25$ units up and k left for what value of k would $f(x)$ be an odd function (1 Mark)

The **goons** want the bridge to contact Mr. Kermonds Ferrari when it reaches the local maximum of $f_2(x)$, they also want the bridge to subtend an angle of 45 degrees with the tangent to $f(x)$ at $x = 1$

d. State the equation of the bridge (2 Marks)

Mr. Kermond realizes he was wrong about the model, he wonders why his mathematical acumen is so weird today

Section B of the Lucas Leap Highway is actually modelled by the function

$$k(x) = (\sqrt{x - c})^{-1} + (\sqrt{nc - x})^{-1}$$

A glitch in existence (an asymptotes occurs $c=6$ and the section ends

f.)

i.) **State** the value of c and the value of n (1 Mark)

ii.)

Use the value you found in part i.) for n

For what value of c does the tangent line at $x=3$ pass through the point $(0, 2\sqrt{c})$
(1 Mark)

Use this value of c (i.) for the remainder of the question

g.)

i.) The Domain of $k(x)$ where c is the value found in f.) ii.) is restricted such that $a < x < 18$

Find the smallest value of a , such that the inverse function of $h(x)$ exists (1 Mark)

ii.)

Find the value(s) of n so that tangent $k(x)$ is tangent to $y = 2x - 1$ (2 Marks)

The Final Section of the highway Section C can be modelled by a function in the form

$$f_3(x) = \frac{a}{x-h} + k$$

It is known that 1800 m to the origin there is a near infinite glitch in reality in the highway, it is further known that $f_3(x)$ is an involution, meaning that it is its own inverse function,

h.

i. Find the values of h and k

(2 Marks)

ii. The equation $x/2 - 1$ intersects at least twice with $f_3(x)$, state the value(s) of a (2 Marks)

a^-

Mr. Kermond and Will exit the highway and arrive in Mt. Grainger

***“WOOOOOOOOOOOOOOF WOOOOOOOOOOOOOOOOOOOOOOOOOOOOOOOOOOF
WOO”***

Finchy barked (even though he can talk he likes barking sometimes)

“We Must find the Schmiddler, he could be anywhere in this town”

- Mr. Kermond declared

“Wait a second , who’s that” - William asked Mr. Kermond, pointing to an old woman at the sidewalk of the entrance to Mt. Grainger

“Well hello there my friends, weary travellers - Cackled the woman in an raspy, yet soothing voice

“And who exactly are you Ma’am”-Mr. Kermond jeered sharply

“I am Mrs. Sings witch of the highway “

“Do you belong to some sort of old people’s home, I might go there after retiring from my Mathematics teaching”

“Now now now let’s not be so hasty , I just wanted to direct you to a most wondrous food court in Mt. Grainger it’s called - Path of shawarma, best indian food in town”

“Wait a second, Mr. Kermond, that’s where the Schmiddler was recorded to be!!!”

“Your right Will”

“Alright thanks lady we’ll head there right now”

“OOOOO Yes” - The witch Cackled once again,

“Mission Control ,heading over to the Path of Shawarma”- Mr. Kermond commanded

The gang arrived at the Path of Shawarma , and their senses were immediately overhauled , the sweet aroma of south asian spices, invaded their olfactory capacity whilst the art painted on the walls gave their eyes a feeling of salubrious empowerment

“Ah welcome welcome” - The owner whispered in a calm, tempered voice

“Um, hello, who are you?”- William instinctively pondered

“I am Parth Singh the owner of this establishment”

“I used to be a maths teacher and astrophysicist but now I found greater delight in the culinary arts , I made this food court to satisfy my entropy with life”

“ Quite a poet you are . Mr. Singh” - Mr. Kermond uttered

“Oh yes I always believed that a writer was an artist and so choose to make art in whatever form they choose”

“May we have a table?”- Enquired Mr. Kermond

“Definitely sir, you can have this table right here

Question 3 (15 Marks)

MISSION STATUS: STANDBY

TIME: 2:00 PM (GMT+11)

MT. GRAINGER, NEAR THE PATH OF SHAWARMA PRECINCT

The Path of **Shawarma** is one of the most famous food courts in all of Mt. Grainger, this is due to a secret ingredient X that's prevalent in all of their foods, little did the gang know, this ingredient is actually very dangerous to the human immune system

Agent Finchy orders the **Triple Exponential Shawarma**

As soon as the mission is logged Finchy eats the Shawarma and the concentration of the secret Ingredient varies with time according to the function

$$X(t) = 3te^{-3t} - t/3$$

Where t is the time in minutes since consumption and X is the concentration in mm/L of blood

a.)

i.) State the implied domain of $X(t)$ (1 Mark)

ii.) State the concentration of Chemical X in Finchy's bloodstream at 2:01 PM, correct to two decimal places (1 Mark)

iii.) **Based** on data of Finchy's immune system, Chemical X is considered poisonous if it's above 0.3 mg/L of blood for more than 150 seconds.

Show that, using relevant calculations the initial bite was not poisonous

(2 Marks)

"Darnit"- Parth screams in anger

Agent Finchy's immune system manages to resist the poison fully and Mr. Kermond attempts to fight Mr. Singh in alarm,

"Hahahaha, don't you even try!!!"

Mr. Singh quickly forcefeeds Triple Exponential Shawarmas laced with Chemical X on William, Mr. Kermond and Finchy again

The Chemical X concentration in Mr. Kermonds' bloodstream evolves according to:

$$X_3(t) = 3te^{(-3t)}$$

After 1 second the poison is again injected into his bloodstream

b.

i.) Determine the maximum concentration of chemical X in his blood stream (1 Mark)

ii.) If his blood is laced with Chemical X again k seconds after the initial dose, what would be the value of k so that the highest concentration is maximized (2 Marks)

The concentration of poison evolving in Will's bloodstream is different and hard to determine, but Mr. Singh knows the function is of the form

$$Xn(t) = (n^2 - kn + 3)te^{(-nt)}$$

c.)

i.) **Show** using algebra, that the maximum concentration always occurs at $t = 1/n$ where $n > 0$

(1 Mark)

ii.) State two transformations, such that the maximum concentration of the poison would be equivalent to $2/e$ mg/L of blood and occurs at $60n$ seconds (2 Marks)

"These fools have fall into my trap"-Mr. Singh cackled in a sardonic tone

"Will-" Mr. Kermond blurted with a spared breath

"I remember obtaining an antidote to the poison it's in my bag-" - He coughs

William, with his last strength manages to strangle Mr. Singh , who was holding Mr. Kermond's collection of items, he grabs the antidote and injects it in himself

The antidote scales down the concentration according to the function

$$An(t) = X(t)e^{-t}$$

Where t is the time in minutes and X is the concentration of Chemical X .

The antidote is applied at 2:03 PM

William will be cured of the concentration if it is under 0.3 Mg/L

d.)

State a relationship between n and k such that William gets cured exactly 10 seconds after the antidote is applied (2 Marks)

Once William cures himself and Mr. Kermond, they see Agent Finchy has disappeared . Mr. Singh has left and a warning system says the Building will explode in k seconds. William managed to snatch a device from Parth

The device contains a cylindrical pipe containing Chemical X with an insulation cover that controls heat transfer - the thermal energy built up results in device triggering an alarm resulting in the bomb's detonation
As shown in the diagram

Mr. Kermond quickly calls his workmate Alex "Agent G." Garrylescu to explain the thermodynamics of the situation . He says in a gruff romanian accent:

"The Equation of Heat transfer in this case can be modelled by

$$Q(R) = \frac{1}{\ln(R) + \frac{C}{R}}$$

Where Q is the *rate* of thermal energy release and R is the **ratio** of The total radius of the pipe with insulation and the inner radius without insulation

"

Let x be the radius of the outer insulation component in metres
It is known the inner radius of the pipe is 300 cm

e.)

i.) Find x in terms of C in which the maximum rate of heat transfer occurs (2 Marks)

The exposure to air results in Chemical X burning the insulation layer at a rate of $5 \text{ m}^2/\text{s}$, when the heat transfer reaches 200 W the bomb will burst

ii.) Find the minimum value of C so that the bomb does not burst.

William manages to cure himself and Mr. Kermond, and they leave the building

"hMPh! , What do we do now , said Mr. Kermond,

They attempt to return to Mr. Kermonds' Ferrari, however, they find it being stolen by some goons in black robes

"Hey what do you think you're doing boneheads!!!"

They once again see the old woman in distance, Mrs. Sings

"YOU OLD HAG" - Mr. Kermond bellows

"Want to know where they're going? go to this address"

Mrs. Sings gives them a parchment , inscribed on it in Comic Sans font was the address of an abandoned circus

Mr. Kermond begrudgingly took it and and they saunter to the address



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SD-HBA-GDTR

Question 4 (10 Marks)

MISSION STATUS: ONGOING

TIME: 4:00 PM (GMT+11)

Mr. Mays' circus of wonders-The Big wheel (Mt. Grainger)

Mr. Kermond and Will arrive at the address written by Mrs. Sings.

It was an old circus named "Mr. Mays' Circus of wonders"

"Huh what is this " Mr. Kermond says

At the old circus, they see a large ferris wheel, shown in the diagram below



It is known the ferris wheel has a diameter of 50 metres (m)

As soon as Will and Mr. Kermond arrive at Mr. Mays' circus, the wheel suddenly starts rotating in an anticlockwise direction at an angular speed of 25 radians per second

The height of the carriage that carries his Ferrari relative to the origin at the centre of the circle

Can be modelled by the following motion equation

$$R(t) = A \cos(B(t - c)) + D$$

Where $0 < c < \frac{\pi}{24}$ and t is the time in seconds and R is the height in meters

a.

i.) Find the values of A B c and D

(2 Marks)

ii.) State the vertical velocity of the carriage at 4:03 PM

(1 Mark)

iii.) Show using algebra that $R(t)$ in the form $q + r \sin(c - bt)$ where q , r , and b are integers (1 Mark)

According to what Mrs. Sings, Witch of the Highway said a person once fell off from the ferris wheel from it's highest height , and now haunts the circus due to it's lack of safety regulations .

b.)

Someone is said to be haunted by the ghost when the position of the carriage is more than 10 m below the highest height

Find the duration of time to the nearest two seconds in the first cycle of the ferris wheel, that a traveller is said to be haunted by the ghost (2 Marks)

c. Let H be the height relative to the center of the wheel that Mr. Kermond's ferrari is above for 25% of the time, find the value of H (2 Marks)

Suddenly one of the **goons** begins to tamper with the controls governing the angular velocity of the ferris wheel

d.

The ghost will be summoned if the carriage traces an arc that subtends an angle of atleast 45 degrees in 10 seconds

i.) State the minimum value of B of the wheel's rotation, such that the ghost is summoned , furthermore , if this value is used state a transformation needed to maintain initial conditions. (2 Marks)

ii.) **What** velocity would satisfy the conditions such that the arc length of the trace is in 10 seconds is equal to the radius of the wheel

(1 Mark)

The tampering results in a chain reaction as the ghost exerts an ethereal dual tone frequency signal, the the strength of the signal can be modelled by the sum of two signals as:

$$T(t) = A \sin(b * t) + C * \cos(d * t)$$

Where t is the time in seconds

e.)

i.) State in terms of b and d , assuming b and d are co-prime integers , the smallest possible period of the dual signal (1 Mark)

ii.) If $b = d$, **Show** that , using algebra that in terms of A and C the maximum possible value of T(t) is $\sqrt{A^2 + C^2}$

(1 Mark)

iii.) **Show** that if $b = d$, using Algebra , that the maximum value of T(t) occurs at $t = \tan^{-1}(A/b)$

(1 Mark)

f.)

A electromagnetic sensor in the ferrari of the dashboard has shown that the frequency of one frequency (that is modelled by the sine function oscillates two times faster than the other .

i.) **State** the equation of $T(t)$ in terms of only b (1 Mark)

ii.) If $A = C$ State the minimum value of $T(t)$ (1 Mark)

- iii.) Now , let the frequency of the first signal be k times faster than the second
Assuming $A = C$, what is the minimum possible maximum value of $T(t)$
(2 Marks)

Mr. Kermond manages to get the ferrari back but the ghost traps them in an electromagnetic field , the goons surround at them gunpoint forcing them both to stand down

***Suddenly a hooded figure, looms over the pair
"Muhahahahahah , after all these years John , you still fall into my traps"
"What the- Schmiddler!!!"***



“Schmiddle me this , schmiddle me that!!!”

The Schmiddler is one of Mr. Kermonds worst enemies from his schooldays,

He reveals a safe near the big wheel, showing Agent Finchy Tied up, the goons have their guns aimed at him

“Oh deary me, look at what is going to happen” - Muhahahahaha

“Dont you touch him “- Mr. Kermond declared

“And what are you going to do about, kermit?”

No one called Mr. Kermond Kermit ever since he was a boy, it was a deeply personal insult to him, that he could not tolerate.

The goons attempted to shoot Finchy but Mr. Kermond used an on-board device that he had invented to knock out all the goons due to their metal wear

“Bah” Still with your classic tricks Mr. Kermond?

“Take me on”

Question 5 (6 Marks)

“ You shall not harm me or any of my acquaintances”- Mr. Kermond had cried

“Oh don’t worry i’ll make sure you all meet together, in **heaven**” - The schmiddler replied coldly

In order to fight the Schmiddler, William and Mr. Kermond have to identify two unknown functions $f(x)$ and $g(x)$ using their discovery of it’s properties

With his mathematical acumen and analyzing Schmiddler’s behaviour

Mr Kermond manages to visualize a table of values for the two functions, the values of $f(0)$ and $g(0)$ are not known

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
-3	9	-10	4	?
-2	3	-8	3	?
-1	8	6	2	?
0	a	b	1	1

1	3	6	?	2
2	8	8	?	3
3	9	10	?	4

Using his knowledge of Schmiddler's trickery he declares

"The functions are continuous and have no stationary points outside of this domain"

Let $h(x) = f(x)/g(x)$

a. State the value of $h'(-2)$

(2 Mark)

b. Find the value of $h'(2)$

(1 Mark)

Define $h_2(x) = f(g(x))$

c. If $g(x)$ had m roots and $f(x)$ had n roots, what would the be maximum possible number of roots, in terms of m and n of $h_2(x)$, assume the composition is defined (1 Mark)

Let $h_3(x) = h_2(x) * h(x)$

D.

i.) Find the tangent line to h_3 at $x = 3$

(1 Mark)

Consider $H(x) = 2 * h_3(x/2) + 3$

ii.) State the equation of the tangent to H at $x = 6$

(2 Mark)

The schmiddler fears that Mr. Kermond will be able to solve the final problem :
In order to stop him he suddenly sprays him with a special variant of Chemical X
It's now up to William Patel to solve the final problem in order to make sure he doesn't pass away

Let $k(x)$ be the inverse of $f(x)$

Let $l(x)$ be the inverse of $g(x)$

Let $m(x) = k(l(x))$

e. Find $m'(3)$

(1 Mark)

The Schmiddler is surprised by William's ability to decipher the properties of the functions, and the ghost- which turned out be to be an illusion by the schmiddler

The field breaks and the ghost disappears, William's immune system suddenly adapts to the concentration of Chemical X, his cells are radioactively altered, give him powers that knows not of yet

"What do we do now sir," Asked William

"We must return to JKSS , this situation is dire"

Will the gang be able to recover the SAC before it's too late? find out soon

TO BE CONTINUED

**END OF VCE
MATHEMATICAL METHODS SAC/IA PART I
STAY TUNED FOR PART II!!!**

EXTRA WORKING: